

Online Appendix to

Counterfactual Sufficiency

State Representation and Response Geometry

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This appendix contains proofs, simulation design, and quantitative implementation details for the paper. It follows the notation, target definitions, and state-design contracts used in the main text.

Appendix A. Proofs

A.1. Proof of Theorem 1

PROOF. If $z(s) = z(s')$, the first coordinate gives $q(s) = q(s')$ and the factorization $\mathbf{R}_{\mathcal{T}} = g \circ z$ gives $\mathbf{R}_{\mathcal{T}}(s) = \mathbf{R}_{\mathcal{T}}(s')$. Hence s and s' belong to the same counterfactual-equivalence class. Define $h(z(s)) = \pi_{\mathcal{T},q}(s)$. The preceding implication makes h well defined and gives $\pi_{\mathcal{T},q} = h \circ z$ on the attained image. For the local statement, let $V = \ker Dq(s)$ and $A = D\mathbf{R}_{\mathcal{T}}(s)|_V$. Write $C = Db(s)|_V$. First-order closure implies $\ker C \subseteq \ker A$. Rank-nullity gives $\text{rank } C = \dim V - \dim \ker C \geq \dim V - \dim \ker A = \text{rank } A$. Since $\text{rank } C \leq d$, $d \geq \text{rank } A$. For attainment, choose a complement $V = \ker A \oplus W$ with $\dim W = \text{rank } A$. Define a linear map $C^* : V \rightarrow \mathbb{R}^{\text{rank } A}$ that vanishes on $\ker A$ and is an isomorphism on W . Extend C^* to the full tangent space and realize the extension as the derivative at s of a C^1 map in a local chart. Then $\ker Dq(s) \cap \ker Db(s) = \ker A$, which gives first-order closure. $\square$

A.2. Proof of Proposition 1

PROOF. For $v \in \mathbb{R}^k$,

$$(A1) \quad v' H_{\mathcal{T}}(s) v = \sum_{t \in \mathcal{T}} \omega_t \|W_t^{1/2} A_t(s) v\|_2^2.$$

Every summand is nonnegative and each ω_t is positive. Hence the quadratic form is zero exactly for vectors in every $\ker\{W_t^{1/2} A_t(s)\}$. The stacked weighted map $A_{\mathcal{T},W}$ has Gram operator $H_{\mathcal{T}}$, so they have the same kernel and rank. If W_t is positive definite on $\text{im } A_t(s)$, then $W_t^{1/2} A_t(s) v = 0$ exactly when $A_t(s) v = 0$. Frame equivariance follows by substituting $\tilde{A}_t = A_t O$. $\square$

A.3. Proof of Proposition 2

PROOF. Write a vertical vector as $Zz = ZC\tilde{z}$. The pullback metric gives $\tilde{G} = C'GC$. Each target derivative in the new coordinate is $A_t C$, so its quadratic contribution is $C' A_t' W_t A_t C$ and $\tilde{Q} = C' Q C$. If $Qv = \lambda Gv$ and $v = C\tilde{v}$, left multiplication by C' gives $\tilde{Q}\tilde{v} = \lambda \tilde{G}\tilde{v}$. Hence the generalized spectrum is invariant and the tangent direction represented by v is the same geometric vector as the direction represented by $\tilde{v}$. For target coordinates satisfying $y = D\tilde{y}$, the identity $y' W y = \tilde{y}' D' W D \tilde{y}$ gives the stated metric transformation and leaves the target quadratic form unchanged. $\square$

A.4. Proof of Theorem 2 and Corollary 1

PROOF. For each target view, $L_t = W_t^{1/2} A_t$ is Hilbert–Schmidt and $A_t^* W_t A_t = L_t^* L_t$ is positive trace class with $\|L_t^* L_t\|_1 = \|L_t\|_{\text{HS}}^2$. Condition (20) therefore gives Bochner

integrability in the trace-class Banach space. The integral in (21) is positive and self-adjoint, and the trace identity (22) follows from continuity of the trace on positive trace-class operators. Trace-class operators are compact. For $v \in \mathbb{V}_s$,

$$(A2) \quad \langle \mathfrak{H}_{\mathcal{T},s} v, v \rangle_s = \int_{\mathcal{T}} \omega_t \|W_t^{1/2} A_t v\|_{\mathbb{V}_t}^2 dv(t).$$

The integrand is measurable and nonnegative. The integral is zero exactly when $W_t^{1/2} A_t v = 0$ for v -almost every target. Positive definiteness of W_t on the target image gives (23). For the sieve statement, let $\Pi_m \rightarrow I$ strongly. A trace-class operator admits finite-rank approximants in trace norm. For a finite-rank operator F , strong convergence of Π_m is uniform on the finite-dimensional range of F and F^* , which gives $\|F - \Pi_m F \Pi_m\|_1 \rightarrow 0$. Given $\epsilon > 0$, choose finite-rank F with $\|\mathfrak{H} - F\|_1 < \epsilon$. Since orthogonal projections are contractions,

$$(A3) \quad \|\mathfrak{H} - \Pi_m \mathfrak{H} \Pi_m\|_1 \leq \|\mathfrak{H} - F\|_1 + \|F - \Pi_m F \Pi_m\|_1 + \|\Pi_m (F - \mathfrak{H}) \Pi_m\|_1$$

$$(A4) \quad \leq 2\epsilon + \|F - \Pi_m F \Pi_m\|_1.$$

Taking m to infinity and then ϵ to zero gives (24). Trace-norm perturbation bounds for self-adjoint compact operators give (25), and operator-norm convergence follows from $\|\cdot\|_{\text{op}} \leq \|\cdot\|_1$. A separated spectral projector therefore converges by the Davis–Kahan theorem. Ky Fan’s principle gives the eigenvalue-tail loss. For Corollary 1, write an absolute perturbation as $h = \mu s$. The Fisher pairing becomes $\mathbb{E}_\mu[ss']$. The derivative of the supplied moment $\mathbb{E}_\mu[\phi_j]$ in direction s is $\mathbb{E}_\mu[\phi_j s]$, which gives (26). The Riesz representation of the restricted target derivative is the orthogonal projection $\psi_t^\vee$. A scalar target therefore contributes the rank-one operator $\psi_t^\vee \otimes \psi_t^\vee$, giving (27). For a finite family, collect the score gradients as columns of an operator $\Psi : \mathbb{R}^M \rightarrow \mathbb{V}_\mu$. The nonzero eigenvalues of $M^{-1} \Psi \Psi^*$ equal those of $M^{-1} \Psi^* \Psi$, which is the view Gram matrix. $\square$

A.5. Proof of Theorem 3

PROOF. Let $\mathcal{A}$ be the stacked weighted operator with block rows $\sqrt{\omega_t} W_t^{1/2} A_t$. Then $H_{\mathcal{T}} = \mathcal{A}' \mathcal{A}$. For a rank- d orthogonal projector P , isotropic quadratic loss from discarding $(I - P)v$ equals $\text{tr}\{\mathcal{A}(I - P)\mathcal{A}'\} = \text{tr}[(I - P)H_{\mathcal{T}}]$. The Ky Fan variational principle maximizes $\text{tr}(PH_{\mathcal{T}})$ over rank- d projectors at the leading eigenspace, giving (29). Worst-case norm loss over $\|v\| \leq \rho$ equals $\rho \|\mathcal{A}(I - P)\|_{\text{op}}$. The Eckart–Young singular-value characterization gives the minimum $\rho \sigma_{d+1}(\mathcal{A}) = \rho \sqrt{\lambda_{d+1}(H_{\mathcal{T}})}$. Equation (31) follows from the eigenvalue tail. $\square$

A.6. Proof of Proposition 3

PROOF. Suppose the coordinate map exists. Smooth vector fields X and Y taking values in $\mathcal{K}_d$ annihilate every component f of (q, b) . The identity $[X, Y]f = X(Yf) - Y(Xf) = 0$ gives $[X, Y] \in \ker D(q, b) = \mathcal{K}_d$, so $\mathcal{K}_d$ is involutive. Conversely, the constant-rank theorem gives local coordinates (x, y) with $q(x, y) = x$ and vertical space spanned by the y coordinates. Frobenius gives adapted vertical coordinates (b, w) on a smaller neighborhood in which $\mathcal{K}_d$ is spanned by the w directions. In the combined coordinates (x, b, w) , the differential kernel of (q, b) is $\mathcal{K}_d$. $\square$

A.7. Proof of Proposition 4

PROOF. Add and subtract the leading spectral projector:

$$(A5) \quad \text{tr}\{(I - P_b)H_{\mathcal{T}}\} = \text{tr}\{(I - P_d)H_{\mathcal{T}}\} + \text{tr}\{(P_d - P_b)H_{\mathcal{T}}\}.$$

The first term is the spectral tail. Ky Fan's principle makes the second term nonnegative. Under an eigengap, equality in the rank- d Ky Fan problem identifies $P_b = P_d$. Trace duality gives (50). Integration gives (51). $\square$

A.8. Proof of Theorem 7

PROOF. Conditional expectation is the orthogonal projection of $\mathbf{F}_{\mathcal{T}}(S)$ onto the closed L^2 subspace measurable with respect to $z(S)$. Apply (54) to every coordinate of $\mathbf{F}_{\mathcal{T}}$ in an orthonormal output basis and sum. The tangent space of a regular level set is $\ker Dz(s)$. Integration over the disintegration of μ gives (55). For $z = (q, b)$,

$$(A6) \quad \|\mathbf{D}\mathbf{F}_{\mathcal{T}}(s)\Pi_{\ker Dz(s)}\|_{\text{HS}}^2 = \text{tr}\{(I - P_b(s))H_{\mathcal{T}}(s)\}.$$

Proposition 4 gives (56). Alignment and the uniform bound on C give (57). $\square$

A.9. Proof of Theorem 8 and Proposition 6

PROOF. The Hilbert-space conditional-variance identity gives

$$(A7) \quad \mathfrak{R}(z) = \mathbb{E}\|\mathbf{F}_{\mathcal{T}} - \mathbb{E}[\mathbf{F}_{\mathcal{T}} | z, C_z]\|_2^2$$

$$(A8) \quad + \mathbb{E}\|\mathbb{E}[\mathbf{F}_{\mathcal{T}} | z, C_z] - \mathbb{E}[\mathbf{F}_{\mathcal{T}} | z]\|_2^2.$$

Apply the componentwise conditional Poincaré inequality to the first term. The second term equals the final term in (60). Vanishing total risk is equivalent to almost-sure factorization through z . For the finite-state result, assign a distinct real number to every class

of the finite exact target quotient and define $b(s)$ as the label of the class containing s . Equality of (q, b) implies equality of the target class. $\square$

A.10. Proof of Theorem 9

PROOF. Let $X_1, \dots, X_d, Y$ be independent and uniform on $[-1, 1]$, let q be constant, put $\phi_k(u) = \cos\{k\pi(u+1)/2\}$, fix $A > 1$, and set

$$(A9) \quad F(X_1, \dots, X_d, Y) = A \sum_{j=1}^{d-1} \phi_2(X_j) + \epsilon \phi_N(X_d) + \phi_1(Y),$$

where the sum is empty when $d = 1$. Choose an even integer N . Since $\mathbb{E}\phi'_k(U) = 0$ for even k and U uniform on $[-1, 1]$, independence removes every cross term. Apart from the common factor $\pi^2/8$, the integrated derivative operator is diagonal with entries $4A^2$ for $X_1, \dots, X_{d-1}$, $\epsilon^2 N^2$ for X_d , and 1 for Y . Choose $\epsilon > 0$ such that $1/\epsilon^2 \geq K$, and then choose the even N large enough that $\epsilon^2 N^2 > \max\{1, \rho/(1-\rho)\}$. The leading d -dimensional eigenspace is therefore spanned by $X_1, \dots, X_d$, and its derivative share is

$$(A10) \quad \frac{4(d-1)A^2 + \epsilon^2 N^2}{4(d-1)A^2 + \epsilon^2 N^2 + 1} > \rho.$$

Compare $b_{AS} = (X_1, \dots, X_d)$ with $b_{alt} = (X_1, \dots, X_{d-1}, Y)$. Independence gives

$$(A11) \quad \Re(q, b_{AS}) = \text{Var}\{\phi_1(Y)\} = \frac{1}{2}, \quad \Re(q, b_{alt}) = \text{Var}\{\epsilon \phi_N(X_d)\} = \frac{\epsilon^2}{2}.$$

Hence $\Re(q, b_{AS})/\Re(q, b_{alt}) = 1/\epsilon^2 \geq K$. $\square$

A.11. Proof of Theorem 10

PROOF. Fix a connected component of a regular fiber and suppress its indices. Let $\{\phi_k\}_{k \geq 0}$ be an orthonormal eigenbasis of $\mathcal{L}$, with $\mathcal{L}\phi_k = \nu_k \phi_k$ and $0 = \nu_0 < \nu_1 \leq \nu_2 \leq \dots$. Each scalar coordinate r_ℓ of the conditional target residual has mean zero and an expansion $r_\ell = \sum_{k \geq 1} a_{\ell k} \phi_k$. The conditional within-component risk, derivative certificate, and slack diagnostic are

$$(A12) \quad V = \sum_{\ell, k \geq 1} a_{\ell k}^2,$$

$$(A13) \quad C = \sum_{\ell, k \geq 1} \frac{\nu_k}{\nu_1} a_{\ell k}^2,$$

$$(A14) \quad D = \sum_{\ell, k \geq 1} \left(\frac{\nu_k}{\nu_1} - 1 \right)^2 a_{\ell k}^2.$$

The Dirichlet-form identity gives the middle expression. Therefore $C - V \geq 0$, and Cauchy-Schwarz gives

$$(A15) \quad C - V = \sum_{\ell, k \geq 1} \left(\frac{\nu_k}{\nu_1} - 1 \right) a_{\ell k}^2 \leq \sqrt{VD}.$$

Integrate over fiber values and component labels. The expansion for $C - V$ gives (65). Applying Cauchy-Schwarz in the direct-integral conditional L^2 space gives (66). The integrated within-component variance is $\Re(q, b) - \Gamma(b)$, while the between-component term enters both $\mathcal{C}^\sharp(b)$ and $\Re(q, b)$ without approximation. The difference is zero exactly when every coefficient with $\nu_k > \nu_1$ vanishes. For a nonzero residual, equality in Cauchy-Schwarz holds exactly when $\mathcal{A}_b r_b = \kappa r_b$ for one nonnegative constant κ in that direct-integral space. $\square$

A.12. Proof of Proposition 5

PROOF. Fix a cell g and suppress the candidate and cell indices. Because the graph is connected and L is nonnegative and self-adjoint in the empirical inner product, it has an orthonormal eigenbasis $\{\phi_k\}_{k=0}^{n_g-1}$ with $L\phi_k = \nu_k \phi_k$ and $0 = \nu_0 < \nu_1 \leq \dots$. Each centered residual coordinate has an expansion $r_\ell^\circ = \sum_{k \geq 1} a_{\ell k} \phi_k$. Its contribution to certificate minus risk and to the diagnostic is

$$(A16) \quad \sum_{\ell, k \geq 1} \left(\frac{\nu_k}{\nu_1} - 1 \right) a_{\ell k}^2 \quad \text{and} \quad \sum_{\ell, k \geq 1} \left(\frac{\nu_k}{\nu_1} - 1 \right)^2 a_{\ell k}^2.$$

The first expression is nonnegative. Cauchy-Schwarz bounds it by the square root of the centered residual norm times the diagnostic. Weighting by π_g , summing across cells, and applying Cauchy-Schwarz once more gives (70)–(71); the cell-mean term enters risk and certificate equally. Conditional on the auxiliary sample, Hoeffding's inequality bounds the deviation for one candidate by $M_G \sqrt{\log(2|\mathfrak{B}_{d,G}|/\alpha)/(2n_G)}$. A union bound over the finite menu gives (72). $\square$

A.13. Proof of Theorem 11

PROOF. Put $P_n f = n^{-1} \sum_{i=1}^n f(S_i)$ and $Pf = \mathbb{E}f(S)$. When $L = 0$, all ψ_η agree almost surely and a single application of Hoeffding's inequality is bounded by (75). Suppose $L > 0$. Take an r -net of Ξ_d . Its cardinality is at most $(1 + 2D_\Xi/r)^{p_d}$. Hoeffding's inequality and a union bound give, with probability at least $1 - \alpha$,

$$(A17) \quad \max_{\eta \text{ in the net}} |(P_n - P)\psi_\eta| \leq M \sqrt{\frac{p_d \log(1 + 2D_\Xi/r) + \log(2/\alpha)}{2n}}.$$

The Lipschitz condition extends this bound from the net to Ξ_d at cost $2Lr$. Setting $r = M/(2L\sqrt{n})$ gives

$$(A18) \quad \sup_{\eta \in \Xi_d} |(P_n - P)\psi_\eta| \leq a_n(\alpha).$$

Consequently,

$$(A19) \quad e_n := \sup_{\eta \in \Xi_d} |\widehat{\mathfrak{C}}_n(\eta) - \mathfrak{C}^\sharp(b_\eta)| \leq \rho_n + a_n(\alpha).$$

The certificate sandwich and approximate minimization imply

$$(A20) \quad \mathfrak{R}(q, b_{\widehat{\eta}_d}) \leq \mathfrak{C}^\sharp(b_{\widehat{\eta}_d})$$

$$(A21) \quad \leq \widehat{\mathfrak{C}}_n(\widehat{\eta}_d) + e_n$$

$$(A22) \quad \leq \widehat{\mathfrak{C}}_n(\eta_d^*) + \eta_n + e_n$$

$$(A23) \quad \leq \mathfrak{C}^\sharp(b_d^*) + \eta_n + 2e_n$$

$$(A24) \quad \leq \mathfrak{R}_d^* + \sqrt{\{\mathfrak{R}_d^* - \Gamma(b_d^*)\}\mathfrak{D}(b_d^*)} + \eta_n + 2\rho_n + 2a_n(\alpha).$$

The lower bound follows from (61). The score envelope and the certificate sandwich give $\mathfrak{R}_d^* \leq M$, so the uniform bound on $\mathfrak{D}$ gives the final assertion. $\square$

A.14. Proof of Corollary 2

PROOF. Intersect the empirical-process event in (A18) with the two events in (78). A union bound gives probability at least $1 - \alpha - \beta_{1n} - \beta_{2n}$. On this intersection, Theorem 11 and $\mathfrak{R}_d^* \leq M$ give

$$(A25) \quad \sqrt{\{\mathfrak{R}_d^* - \Gamma(b_d^*)\}\mathfrak{D}(b_d^*)} \leq \sqrt{M \left\{ \sup_{\eta \in \Xi_d} \widehat{\mathfrak{D}}_n(b_\eta) + s_n \right\}},$$

which proves (79). Intersecting only (A18) and the first-stage score event, then applying the relative-slack condition, proves (80). $\square$

A.15. Proof of Theorem 13 and Corollary 3

PROOF. For any $v \in \mathbb{R}^p$,

$$(A26) \quad v' I_{\mathcal{D}} v = \sum_{e: n_e > 0} n_e \|\Sigma_e^{-1/2} J_e v\|_2^2.$$

Hence (87) follows. The linear-factorization lemma states that linear maps J and T satisfy $\ker J \subseteq \ker T$ exactly when $T = LJ$ for a linear map L on $\text{im } J$. Restrict T_θ to $V = \ker J_{\mathcal{D}}$. If C is the derivative of d added scalar moments and target identification holds after augmentation, then $\ker(C|_V) \subseteq \ker(T_\theta|_V)$. Rank-nullity gives $d \geq \text{rank}(T_\theta|_V)$. A map with kernel equal to $\ker(T_\theta|_V)$ and image dimension $r_{\mathcal{D} \rightarrow \tau}$ attains the count. The rank formula in (89) is the rank increment from stacking T_θ below $J_{\mathcal{D}}$. Corollary 3 subtracts the two rank increments. Nonnegativity follows from submodularity of row-space dimension. $\square$

A.16. Proof of Lemma 1

PROOF. Expand $\widehat{A}^{(1)'} W \widehat{A}^{(2)}$ as $A'WA + E^{(1)'}WA + A'WE^{(2)} + E^{(1)'}WE^{(2)}$ and take the conditional expectation. The stated moment conditions remove the three error terms. The same-sample identity follows from the expansion $\widehat{A}'W\widehat{A} = A'WA + E'WA + A'WE + E'WE$. $\square$

A.17. Proof of Theorem 14

PROOF. Write $E_{it}^{(b)} = \widehat{A}_{it}^{(b)} - A_t(s_i)$. For one target view,

$$\begin{aligned} \text{(A27)} \quad & \frac{1}{2} \{ \widehat{A}^{(1)'} W \widehat{A}^{(2)} + \widehat{A}^{(2)'} W \widehat{A}^{(1)} \} - A'WA = \frac{1}{2} \{ E^{(1)'}WA + A'WE^{(2)} + E^{(2)'}WA + A'WE^{(1)} \} \\ \text{(A28)} \quad & + \frac{1}{2} \{ E^{(1)'}WE^{(2)} + E^{(2)'}WE^{(1)} \}. \end{aligned}$$

Bounded A_t , W_t , and weights together with (94) give

$$\text{(A29)} \quad \max_i \|\widetilde{H}_i - H_{\mathcal{T}}(s_i)\|_{\text{op}} = O_p(\zeta_n + \zeta_n^2).$$

Next decompose

$$\text{(A30)} \quad \widehat{H}_n(s) - H_{\mathcal{T}}(s) = \sum_i \kappa_{ni}(s) \{\widetilde{H}_i - H_{\mathcal{T}}(s_i)\} + \sum_i \kappa_{ni}(s) \{H_{\mathcal{T}}(s_i) - H_{\mathcal{T}}(s)\}.$$

The first term is uniformly $O_p(\zeta_n + \zeta_n^2)$. Hölder continuity of A_t and boundedness imply Hölder continuity of $H_{\mathcal{T}}$ with the same exponent, so the second term is uniformly $O(h_n^\beta)$ on the support of the partition weights. Combining the bounds gives (99). $\square$

A.18. Proof of Proposition 7

PROOF. The Woodbury identity gives

$$\text{(A31)} \quad (M + J'R^{-1}J)^{-1} = \Sigma - \Sigma J'(R + J\Sigma J')^{-1}J\Sigma.$$

Taking the trace against $H_{\mathcal{T}}$ yields (92). For a scalar view, write $J = a'$ and $R = v^{-1}$. The middle inverse becomes $(v^{-1} + a'\Sigma a)^{-1}$, which gives (93). $\square$

A.19. Proof of Corollary 4

PROOF. Write $\widehat{O}_i = \widehat{O}_{i \rightarrow s}$ and $O_i = O_{i \rightarrow s}$. For each active anchor,

$$(A32) \quad \widehat{O}_i \widetilde{H}_i \widehat{O}_i' - H_{\mathcal{T}}(s) = \widehat{O}_i \{\widetilde{H}_i - H_{\mathcal{T}}(s_i)\} \widehat{O}_i'$$

$$(A33) \quad + \{\widehat{O}_i H_{\mathcal{T}}(s_i) \widehat{O}_i' - O_i H_{\mathcal{T}}(s_i) O_i'\}$$

$$(A34) \quad + \{O_i H_{\mathcal{T}}(s_i) O_i' - H_{\mathcal{T}}(s)\}.$$

The first term has operator norm $O_p(\zeta_n + \zeta_n^2)$ uniformly over active anchors. The second term is bounded by a constant times $\|\widehat{O}_i - O_i\|_{\text{op}}$ because the target operators and orthogonal transports are uniformly bounded. The third term is $O(h_n^\beta)$ by (101). Convex averaging with the partition weights gives (102). $\square$

A.20. Proof of Corollary 5

PROOF. Equation (105) follows from Weyl's eigenvalue inequality. Davis–Kahan applied under Assumption 2 gives (106). Let $E = \widehat{H} - H_{\mathcal{T}}$. Since $\widehat{P}_d$ maximizes $\text{tr}(P\widehat{H})$ over rank- d projectors,

$$(A35) \quad \text{tr}\{(P_d - \widehat{P}_d)\widehat{H}\} \leq 0.$$

Therefore

$$(A36) \quad \mathcal{L}_s(\widehat{P}_d) - \mathcal{L}_s(P_d) = \text{tr}\{(P_d - \widehat{P}_d)H_{\mathcal{T}}\}$$

$$(A37) \quad \leq \text{tr}\{(P_d - \widehat{P}_d)(H_{\mathcal{T}} - \widehat{H})\}$$

$$(A38) \quad \leq \|P_d - \widehat{P}_d\|_* \|E\|_{\text{op}}.$$

The difference of two rank- d projectors has rank at most $2d$, so $\|P_d - \widehat{P}_d\|_* \leq 2d\|P_d - \widehat{P}_d\|_{\text{op}}$. On $\|E\|_{\text{op}} < \gamma_d/2$, the operator-norm Davis–Kahan bound gives $\|P_d - \widehat{P}_d\|_{\text{op}} \leq 2\|E\|_{\text{op}}/\gamma_d$. Combining the inequalities gives (107). For fixed s and separated eigenvalues, the Fréchet derivative of the rank- d spectral projector in direction E is

$$(A39) \quad DP_H[E] = \sum_{j \leq d} \sum_{k > d} \frac{v_k v_k' E v_j v_j' + v_j v_j' E v_k v_k'}{\lambda_j - \lambda_k}.$$

An asymptotically linear expansion for $\widehat{H}(s)$ and the functional delta method applied to (A39) give a Gaussian limit for $\text{vec } \widehat{P}_d(s)$. $\square$

A.21. Proof of Proposition 8

PROOF. For a local parameter displacement u , the first-order field change is $G_H(s)u$. Its weighted squared norm is $u'J_H u$. Positive definiteness gives a positive quadratic separation for every nonzero u . If J_H has rank r , its kernel is exactly the set of parameter directions whose first-order field change has zero weighted norm. Stacking additional moments stacks their derivative maps, and the same quadratic separation criterion gives the joint rank condition. $\square$

A.22. Proof of Theorem 15

PROOF. Theorem 14 and the structural-solver condition imply

$$(A40) \quad \sup_{\theta, s} \|\{\tilde{H}_{\mathcal{T}, \theta, n}(s) - \hat{H}_n(s)\} - \{H_{\mathcal{T}, \theta}(s) - H_0(s)\}\|_F = o_p(1).$$

Uniform boundedness and the uniform law of large numbers then give uniform convergence of the field-distance term in (112) to its population counterpart in (108). Adding uniform convergence of the behavioral criterion yields $\sup_{\theta \in \Theta} |\hat{Q}_n(\theta) - Q(\theta)| = o_p(1)$. Compactness, continuity, and the unique minimizer condition give (113) by the argmin theorem. $\square$

A.23. Proof of Theorem 16

PROOF. Assumption 4 and Slutsky's theorem give joint convergence of $(\hat{\tau}_{n, m_n}, \hat{\mathfrak{H}}_{n, m_n})$ at the root- n scale around $(\tau, \mathfrak{H})$. The map $\kappa_d(H) = \sum_{j=1}^d \lambda_j(H)$ is Hadamard directionally differentiable on self-adjoint Hilbert-Schmidt perturbations at compact H when $\lambda_d(H) > 0$. Let $\lambda_* = \lambda_d(H)$ and decompose the leading spectral space into the eigenspaces strictly above λ_* and the boundary eigenspace. Fan's variational formula gives

$$(A41) \quad \kappa'_{d, H}(G) = \text{tr}(P_{>}G) + \sum_{j=1}^{d-r} \lambda_j(P_{=}GP_{=} |_{\text{im } P_{=}}).$$

The directional delta method applied to $(\tau, H) \mapsto \tau - \kappa_d(H)$ yields (119). When $\lambda_d > \lambda_{d+1}$, the boundary eigenspace is fully retained and the displayed derivative equals $\text{tr}(P_d G)$. For the bootstrap, define the numerical derivative of κ_d at $\hat{H}$ in direction $\mathbb{G}_n^*$ with step ϵ_n . The conditions $\epsilon_n \rightarrow 0$ and $\sqrt{n}\epsilon_n \rightarrow \infty$ place the base-point estimation error below the numerical derivative step. Conditional weak convergence of the bootstrap pair and the numerical directional delta method give conditional convergence of (120) to the directional derivative law. $\square$

A.24. Proof of Corollary 6

PROOF. On the simultaneous coverage event, $U_{\widehat{d},n} \leq \varepsilon$ implies $\ell_{\widehat{d}} \leq U_{\widehat{d},n} \leq \varepsilon$. The probability of this event converges to at least $1 - \alpha$, which gives (123). Under separation and uniform convergence of the bounds, the inequalities defining d_ε hold with the same ordering for the sample bounds with probability approaching one. $\square$

A.25. Proof of Proposition 9

PROOF. For any bounded function $\bar{V}$ on the reduced state, define $V(s) = \bar{V}(z(s))$. Equations (124) and (125) imply that each action-specific Bellman update of V depends on s through $z(s)$. Hence the Bellman operator maps functions that factor through z into the same class. Contraction of the discounted log-sum Bellman operator gives a fixed point in that class, establishing $V_\theta(s, p) = \bar{V}_\theta(z(s), p)$. The logit conditional choice probabilities are functions of the action-specific value indices and therefore also factor through z . $\square$

A.26. Proof of Proposition 10

PROOF. Choose L so that $2^{-\alpha L}$ is of order ε . For fixed L , minimize $\sum_{\ell=0}^L N_\ell C_\ell$ subject to $\sum_{\ell=0}^L V_\ell/N_\ell \leq \varepsilon^2/2$. The Lagrange first-order conditions give N_ℓ proportional to $\sqrt{V_\ell/C_\ell}$ and minimum stochastic work proportional to

$$(A42) \quad \varepsilon^{-2} \left(\sum_{\ell=0}^L \sqrt{V_\ell C_\ell} \right)^2.$$

Under (128), the summand is of order $2^{(\gamma-\beta)\ell/2}$. The geometric sum is bounded when $\beta > \gamma$, grows at rate L when $\beta = \gamma$, and grows at rate $2^{(\gamma-\beta)L/2}$ when $\beta < \gamma$. Substituting $L \asymp \alpha^{-1} \log_2 \varepsilon^{-1}$ gives (130). A single-level estimator uses $O(\varepsilon^{-2})$ draws at level L , each with cost $O(2^{\gamma L})$, giving the stated work rate. $\square$

A.27. Proof of Theorem 17

PROOF. The first-order condition for the multilevel estimator and a mean-value expansion around the exact estimator give

$$(A43) \quad 0 = \nabla Q_n^{ML}(\widehat{\theta}_E) + \widetilde{J}_n(\widehat{\theta}_{ML} - \widehat{\theta}_E),$$

where $\widetilde{J}_n$ is a Hessian evaluated along the segment between the two estimators. The exact first-order condition gives $\nabla Q_n^E(\widehat{\theta}_E) = 0$. Assumption 5 implies $\nabla Q_n^{ML}(\widehat{\theta}_E) = o_p(n^{-1/2})$ and $\widetilde{J}_n \rightarrow_p J$. Invertibility of J gives (135). Adding the exact-estimator expansion gives (136). $\square$

A.28. Representation, finite-radius, decision, and measurement results

PROOF OF THE UNIVERSAL PROPERTY IN THEOREM 1. The maps $\bar{q}$ and $\bar{R}_{\mathcal{T}}$ are well defined because both objects are constant on each equivalence class. Let z carry q and $R_{\mathcal{T}}$. Define $h(z(s)) = \pi^*(s)$. Equality $z(s) = z(s')$ implies equality of q and $R_{\mathcal{T}}$, hence equality of their quotient classes. This makes h well defined on $z(\mathcal{S})$. The identity $\pi^* = h \circ z$ follows by construction, and surjectivity of $z : \mathcal{S} \rightarrow z(\mathcal{S})$ gives uniqueness. $\square$

PROOF OF THEOREM 4. For the upper bound, let $v_1, \dots, v_d$ be the leading right singular vectors of $\mathcal{A}_s^{\mathcal{T}}$ and set $b(v) = (v'_1 v, \dots, v'_d v)$. The truncated singular expansion gives linear approximation error at most $\rho \sigma_{d+1}$, and (34) adds $K_s \rho^2/2$. For the lower bound, restrict any continuous encoder to the radius- ρ sphere in the span of the first $d+1$ right singular vectors. Borsuk–Ulam gives an antipodal pair $v, -v$ with $b(v) = b(-v)$ (Borsuk 1933). The linear target difference has norm at least $2\rho \sigma_{d+1}$. The two remainders reduce the pairwise separation by at most $K_s \rho^2$. A common decoder value for the pair has maximum error at least one half of their separation. The affine case sets the remainder to zero. $\square$

PROOF OF THEOREM 5. The conditional mean is the L^2 -optimal decoder. Put $U_{\perp} = (I - P)U$. Equations (37) and (34) give

$$(A44) \quad F_s(\rho U) - \mathbb{E}[F_s(\rho U) \mid PU] = \rho \mathcal{A}_s^{\mathcal{T}} U_{\perp} + \delta_{\rho},$$

where $\delta_{\rho} = r_s(\rho U) - \mathbb{E}[r_s(\rho U) \mid PU]$. The conditional-variance identity gives

$$(A45) \quad \mathbb{E}\|\delta_{\rho}\|_2^2 \leq \mathbb{E}\|r_s(\rho U)\|_2^2 \leq K_s^2 \rho^4/4.$$

Isotropy yields

$$(A46) \quad \mathbb{E}\|\mathcal{A}_s^{\mathcal{T}} U_{\perp}\|_2^2 = c \operatorname{tr}\{(I - P)(\mathcal{A}_s^{\mathcal{T}})' \mathcal{A}_s^{\mathcal{T}}\} = c \operatorname{tr}\{(I - P)H_{\mathcal{T}}(s)\}.$$

Expansion of the squared norm and Cauchy–Schwarz give (39). The remainder is uniform over rank- d projectors because the residual trace is bounded by $\operatorname{tr} H_{\mathcal{T}}(s)$. Ky Fan’s principle gives (40). If $P_{s,\rho}$ minimizes local risk, uniformity implies

$$(A47) \quad \operatorname{tr}(P_d H_{\mathcal{T}}) - \operatorname{tr}(P_{s,\rho} H_{\mathcal{T}}) = O(\rho).$$

The eigengap inequality

$$(A48) \quad \operatorname{tr}(P_d H_{\mathcal{T}}) - \operatorname{tr}(P H_{\mathcal{T}}) \geq \frac{\gamma_d(s)}{2} \|P - P_d\|_F^2$$

gives (41). $\square$

PROOF OF THEOREM 6. The first-order condition is $D_a L(a^*(r), r) = 0$. Differentiation at r_0 gives (42). Taylor expansion of $a \mapsto L(a, r_0)$ around a_0 gives one half of the action-Hessian quadratic form plus a smaller-order term. Substitution of (42) yields (43). $\square$

PROOF OF THEOREM 12. On a fiber of $M_{\mathcal{D}}$, the raw target derivative has the rank in (82). The joint-rank identity follows from rank-nullity. First-order exact recovery is equivalent to inclusion of the data-fiber tangent space in the raw target kernel. Any k added scalar fields have derivative rank at most k on the data fiber, which gives the exact-recovery lower bound. A basis of the raw target image on the data fiber gives equality at the derivative level. The residual operator is $A'_{\mathcal{T}, W, \mathcal{D}} A_{\mathcal{T}, W, \mathcal{D}}$, so its rank equals $\delta_{\mathcal{T}, W|\mathcal{D}}$ and (83) follows. The same rank argument applied to the weighted target map gives the loss-repair lower bound and derivative-level attainment. Positive definiteness on the attainable residual image equates the raw and weighted kernels and therefore the two deficiencies. After d orthonormal measurement directions are added inside the unresolved fiber, the remaining isotropic quadratic target loss is the trace of $H_{\mathcal{T}|\mathcal{D}}$ on their orthogonal complement. The Ky Fan variational principle gives (84). Under the neighborhood rank conditions, constant-rank coordinates write $M_{\mathcal{D}}(x, u) = x$ and zero raw deficiency gives $D_u \mathbf{R}_{\mathcal{T}} = 0$. Integration along connected local u -fibers defines the local factor. $\square$

Appendix B. Simulation and quantitative design

B.1. Exchangeable-state DDC

Each hidden binary state in group $g \in \{1, 2\}$ follows a two-state Markov chain. Group 1 uses $\Pr(1|0) = 0.12$ and $\Pr(1|1) = 0.88$; group 2 uses 0.18 and 0.82. Conditional independence and exchangeability imply that the next-period count distribution depends on the current hidden configuration through the two current counts. The mileage transition is deterministic: keep advances one bin up to the maximum and replacement resets mileage to zero. This establishes the pushforward condition in Proposition 9.

B.2. Six-dimensional DDC

Each coordinate takes values in $\{-1, 0, 1\}$. Baseline action payoffs are

$$(A49) \quad u_0(z) = -0.10 - 0.45z_1^2 + 0.18z_4,$$

$$(A50) \quad u_1(z) = -1.15 + 1.35z_1 + 0.80z_4 - 0.18z_4^2.$$

Action-dependent transitions shift the first and fourth coordinates. The first policy view adds $0.85[0.95z_2 + 0.18z_2z_1 + 0.12z_5^2]$ to u_1 and the second adds $0.85[0.95z_3 - 0.16z_3z_4 + 0.10z_6^2]$. The discount factor is 0.9. Gradients use second-order finite differences on the three-point

tensor grid. Response surfaces use polynomial features through degree four and a ridge constant 10^{-8} .

B.3. Scattered recovery

For each selected anchor and each of the two counterfactual gradients, two independent Gaussian error vectors with coordinate standard deviation 0.04 are added. Sample sizes are 50, 100, 200, and 400. Each case uses 300 replications. The same-sample comparison uses $\widehat{A}'\widehat{A}$ and the cross-fitted construction uses the symmetrized product of two noisy estimates.

B.4. Two-dimensional multi-index DDC

The state lies on $[0, 1]^2$. Each action has a product transition kernel with action-dependent persistence and drift. The intervention direction is an angle on $[0, 2\pi)$. A finite intervention of magnitude 0.75 enters replacement utility through two state functions and a second-order angular interaction. State grids contain 5, 9, 17, and 33 points per coordinate. Angle grids contain 4, 8, 16, and 64 equally spaced points. The target operator averages outer products of finite-policy CCP-response gradients across the angle grid. Each Bellman solve uses tolerance 10^{-10} .

B.5. Rotating local geometry

The persistent state uses 61 equally spaced points on $[0, 1]$. Four hidden coordinates take values in $\{-1, 0, 1\}$ and remain fixed across Bellman transitions. Keep advances the persistent state by one grid point and replacement resets it to zero. The two finite policy views load on (z_1, z_2) and (z_3, z_4) through angles $1.15\pi x - 0.2\pi$ and $0.85\pi x + 0.1\pi$ with state-dependent amplitudes. Local hidden-state derivatives use second-order finite differences. The scattered experiment samples 40 hidden configurations per anchor and view, adds independent Gaussian derivative noise with standard deviation 0.015, and uses Gaussian state weights. Anchor counts are 6, 10, 16, 24, and 36. Each case uses 200 replications.

B.6. Target-directed angle acquisition

Three hidden coordinates take values in $\{-1, 0, 1\}$ and remain fixed over the Bellman transition. The persistent mileage state has 15 bins. Twenty-four candidate angles use $a(\vartheta) \propto (\cos \vartheta, \sin \vartheta, 0.75 \cos 2\vartheta)'$ with policy magnitude 0.8. Each angle-specific operator is replaced by its leading rank-one component for the design calculation. The target operator is the equal-weight average across the 24 components and is normalized to unit trace. The target posterior-variance criterion uses prior precision $0.2I$ and per-view information scale 3. The random comparison contains 1,000 acquisition sequences.

B.7. One-period heterogeneous-household diagnostic

The household chooses next-period assets on a 31-point grid over $[0, 15]$. Income takes values $(0.5, 1, 1.5)$ and follows (137). Preferences have discount factor 0.96 and coefficient of relative risk aversion 2. The savings choice receives type-I extreme-value smoothing with scale 0.08. The baseline real interest rate is 0.02, the wage is one, and the transfer is zero. The stationary distribution is computed from the baseline transition matrix. The supplied representation records mass, mean assets, and mean income. The vertical basis is obtained from the null space of these three linear functionals. The Fisher metric is represented in that basis by $B' \text{diag}(\mu^{-1})B$. The five policy views are one-period changes to the real interest rate, transfer, or wage. Policy reverts to baseline from period one onward. Target derivatives are computed from the impact transition and powers of the baseline transition. The target family contains three outcomes at five horizons. Each outcome is divided by its baseline aggregate level before target operators are formed.

B.8. General-equilibrium transition experiment

The experiment in Section 8 uses the general-equilibrium transition routine described here. Assets lie on 15 equally spaced points over $[0, 20]$; income follows (137); and preferences use $\beta = 0.94$, relative risk aversion 2, and extreme-value scale 0.8. Production has capital share 0.36 and depreciation 0.08. The stationary capital stock is found by bisection on $[4, 5]$ until 36 bisection steps have been completed. The resulting household transition matrix determines the stationary distribution. For a transition path, the calculation uses a ten-period terminal horizon. A backward household recursion produces period-specific choice probabilities at a proposed capital path. Forward propagation of the distribution produces an implied capital path, and damped fixed-point updates with weight 0.5 continue until the maximum capital-path discrepancy is below 10^{-8} or 100 iterations have been reached. The capital-income package has initial tax rate 0.25 and the labor-income package has initial rate 0.30; both decay by 0.5 per period. Current tax revenue is rebated through a lump-sum transfer. Responses are divided by the baseline level of each outcome before the target vector is formed. The supplied distributional functionals are total mass, aggregate assets, and the masses of the first two income states. If X stacks these four functions and μ_0 is the stationary mass vector, the construction takes the null space of $(\sqrt{\mu_0}X)'$ and rescales it by $\mu_0^{-1/2}$ to obtain the 41-dimensional Fisher-orthonormal basis B . For each standard-normal direction z_j normalized to unit length, the score is Bz_j and the finite-difference radius is

$$(A51) \quad \epsilon_j = 0.08 / \max_i |(Bz_j)_i|.$$

This guarantees nonnegative perturbed masses $\mu_0(1 \pm \epsilon_j Bz_j)$. The calculation uses 46 normalized directions. Central differences produce the 40 target derivatives, and least squares on the normalized directions reconstructs the derivative map. The decision experiment uses 35 training directions and 11 test directions. The two package-response columns form $G(s)$. The disturbance equals negative one half of the sum of the two baseline package responses, and the action penalty is 2×10^{-7} . A finite-difference derivative of the optimal action with respect to $\text{vec } G$ constructs the decision metric. Linear response decoders are fit on the training directions and evaluated on the held-out directions. The greedy dictionary contains asset powers two through four, asset-income interactions, three wealth cells, and income-specific low-wealth cells. These steps generate Table 2 and Figure 2.

B.9. KW97 fixed-model stress test

The stress test in Section 9 uses eight model periods, 31 common solution draws, and full state-space solution without interpolation. The two parameterizations are the distributed `kw_97_extended_respy` and `kw_97_extended` specifications. Five policies change the schooling reward, white-collar wage, blue-collar wage, military wage, and both civilian wages. Common random numbers and aligned solved state spaces are used for every policy. The 6,520 core states each carry four latent types. A fixed split allocates 70 percent of core states to training and 30 percent to testing. For each supplied representation, two training folds estimate local linear policy-response derivatives. Symmetrized cross-products form the cross-fitted operators. The common target metric and state standardization are estimated on training states. Equal-budget methods then use the same quadratic ridge decoder to predict the 55-component finite-policy response vector on held-out core states. The table averages two derivative-estimation runs for each parameterization and reports small, primary, and large supplied representations.

The finite-graph calculation uses the primary supplied representation and the same training normalization, split, targets, coordinate maps, and ridge penalties. A second two-fold split of the training core states forms out-of-fold decoder residuals. Let g index an exact supplied-state cell and let b_i denote the added coordinates of training state i in that cell. For distinct states in a cell, the graph weight is

$$(A52) \quad w_{ij} = \left(1 + \frac{\|b_i - b_j\|_2^2}{h_g^2} \right)^{-1},$$

where h_g is the median positive pairwise distance in the cell. The diagonal weight is zero and $L_g = \text{diag}(W_g \mathbf{1}) - W_g$. The complete positive-weight graph is connected whenever the cell contains at least two distinct coordinates. Singleton cells contribute their

residual-mean term. The calculation evaluates the first positive eigenvalue, risk, certificate, diagnostic, and both sides of (70)–(71).

Candidate selection minimizes the training graph certificate separately by model, estimation run, and dimension. The held-out comparison is evaluated afterward. For $d = 3$, the selected baseline-value candidate is compared with the counterfactual-fiber candidate by resampling the 1,956 test core states with replacement 1,999 times. The percentile intervals in Table 4 condition on the fitted candidates, selected graph rule, structural model, and realized split. The resulting candidate diagnostics and bootstrap summary generate Figure 3.

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